```
from scipy.stats import personr,spearmanr
```

One-sample t-test

```
from scipy.stats import  ttest_1samp
```

```
result = ttest_1samp(x_bar,popmean = 100 )
result
```

```
TtestResult(statistic=np.float64(1.5549631660464456), pvalue=np.float64(0.1949294727553206), df=np.int64(4))
```

```
# population = 100
x_bar = [103,98,105,101,102]    # sample weight of 5 lays

result = ttest_1samp(x_bar,popmean = 100 )

print("t_value",result.statistic)

print("p_value",result.pvalue)
print("df",result.df)



if (result.statistic>result.pvalue):
  print("fail to reject h0")
else:
  print("reject h0")
```

```
t_value 1.5549631660464456
p_value 0.1949294727553206
df 4
fail to reject h0
```

```
from scipy.stats import ttest_1samp

weights = [102, 105, 98, 101, 104]

t_value, p_value = ttest_1samp(
    weights,
    popmean=100
)

print("t-value:", t_value)
print("p-value:", p_value)

alpha = 0.05

if p_value < alpha:
    print("Reject H0")
    print("The population mean is significantly different from 100.")
else:
    print("Fail to reject H0")
    print("There is not enough evidence that the population mean differs from 100.")
```

```
t-value: 1.6329931618554523
p-value: 0.1778078083562211
Fail to reject H0
There is not enough evidence that the population mean differs from 100.
```

```
Start coding or generate with AI.
```

Independent 2 sample t-test

```
def add(*args):
```

```
  total = 0
  for i in args:
    total = total + i
  return total

add(80,85,90,91,87)
433/5
```

```
86.6
```

```
from scipy.stats import ttest_ind


# hypo
# h0 mu0 = mu1
# h1 mu0 != mu1


# 2 independent sample mean from 2 population
drug_a = [90,99,80,95,97]   92.2
drug_b = [80,85,90,91,87]

ttest_ind(drug_a,drug_b)
```

```
TtestResult(statistic=np.float64(1.4270121360253538), pvalue=np.float64(0.19141860861866955),
df=np.float64(8.0))
```

```
from scipy.stats import ttest_ind

class_A = [12, 14, 13, 15, 11]
class_B = [8, 9, 10, 11, 12]

t_value, p_value = ttest_ind(
    class_A,
    class_B,
    equal_var=False
)

print("t-value:", t_value)
print("p-value:", p_value)

alpha = 0.05

if p_value < alpha:
    print("Reject H0")
    print("The two population means are significantly different.")
else:
  print("there is no difference")
```

```
t-value: 3.0
p-value: 0.017071681233782634
Reject H0
The two population means are significantly different.
```

Paired sample T-test

```
from scipy.stats import ttest_rel
```

```
help(ttest_rel)   # fro reference
```

```

before = [80, 82, 78, 85, 90]
after = [78, 80, 77, 82, 87]

ttest_rel(    before,    after )
```

```
TtestResult(statistic=np.float64(5.879747322073337), pvalue=np.float64(0.004181072135640301), df=np.int64(4))
```

```
from scipy.stats import ttest_rel
```

```python
before = [80, 82, 78, 85, 90]
after = [78, 80, 77, 82, 87]

t_value, p_value = ttest_rel(
    before,
    after
)

print("t-value:", t_value)
print("p-value:", p_value)

alpha = 0.05

if p_value < alpha:
    print("Reject H0")
    print("There is a significant difference between before and after.")
else:
    print("Fail to reject H0")
    print("There is not enough evidence of a difference.")
```

```
t-value: 5.879747322073337
p-value: 0.004181072135640301
Reject H0
There is a significant difference between before and after.
```

```
Start coding or generate with AI.
```